Problem Set 1

Starter

1 (E)**.** Let $\mathrm{GL}(n, \mathbb{R})$ be the general linear group, i.e.,

$$\mathrm{GL}(n, \mathbb{R}) \coloneqq \{A \in M_n(\mathbb{R}) : \det A \neq 0\}.$$

Show that it is a group. Moreover, show that the special linear group

$$\mathrm{SL}(n, \mathbb{R}) \coloneqq \{A \in M_n(\mathbb{R}) : \det A = 1\}$$

is a subgroup of $\mathrm{GL}(n, \mathbb{R})$.

2 (E)**.** Let $(G, \cdot)$ be a group. We define an *opposite group* $(G^{\mathrm{op}}, \circ)$ as follows: $G^{\mathrm{op}} \coloneqq G$ and $a \circ b \coloneqq ba$ for all $a, b \in G^{\mathrm{op}}$. Show that $(G^{\mathrm{op}}, \circ)$ is a group.

3 (E)**.** Let G be a group. Show that $xyz = 1$ implies $zyx = 1$ only if G is an abelian group.

4 (E)**.** (a) Show that every subgroup of $(\mathbb{Z}, +)$ is of the form $k\mathbb{Z}$ for some integer k.

(b) Let $a, b \in \mathbb{Z}$. Show that the set

$$a\mathbb{Z} + b\mathbb{Z} \coloneqq \{n \in \mathbb{Z} : n = ar + bs \text{ for some } r, s \in \mathbb{Z}\}$$

is a subgroup of $(\mathbb{Z}, +)$.

(c) Let $a, b \in \mathbb{Z}$ be not both zero. Show that there exists a positive integer $d \in \mathbb{Z}$ such that $d\mathbb{Z} = a\mathbb{Z} + b\mathbb{Z}$. Deduce that

(i) $d = ar + bs$ for some $r, s \in \mathbb{Z}$.;

(ii) d divides a and b;

(iii) if d' divides a and b, then d' divides d.

Note: This d is called the *greatest common divisor* of a and b.

Advanced

5 (L-M)**.** How many ways are there to complete the following grid so that it is the Cayley table of a group?

$*$	e	a	b	c
e				
a				
b				
c				

Here e is assumed to be the identity.

6 (M)**.** Let $\mathrm{GL}(n, p)$ be the group of all $n \times n$ invertible matrices over a field with p elements. Show that

$$|GL(n, p)| = (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1}) = \prod_{k=0}^{n-1} (p^n - p^k).$$

7 (M)**.** If G is a finite group of even order, then G contains an element $a \neq e$ such that $a^2 = e$. Show that the converse is true as well.

8 (H)**.** Let G be a group in which every element other than the identity has order two. Show that G is abelian. Show also that if G is finite, the order of G is a power of 2.